- Intelligent alerting engines
- Configurable discovery rules
- Flexible templates

Open source tools

Now that we have an overview of the technology, protocols, practices and the key drivers for SLA driven network monitoring, let us take a quick look at some of the interesting open source monitoring tools. The generic architecture is seen in Figure 3.

collectd

collectd is a daemon which collects system performance statistics written in plain C for performance and portability. Everything in collectd is done in plugins, and its memory footprint is so small that it runs even on embedded WLAN routers with OpenWRT.

collectd uses a 'data push' model, wherein the data is collected and then pushed to a multi-cast group or server. It also implements a lightweight binary TLV based network protocol that works for data collection even over slow networks. Since data transmission and reception can be configured separately, the resultant models of network operation are - no networking, uni-cast, proxy and multi-cast.

collectd uses an in-memory self-balancing binary search tree to cache values in memory. This approach results in high resolution statistics being processed at scale. The multi-threaded design of collectd allows for multiple plugins to be queried simultaneously. It must be mentioned that influxDB, a time-series database, and Javascript based client-side frameworks like Grafana can also be used to build feature-rich custom dashboards.

Riemann

Another way to visualise the network is in terms of events raised by the various network elements, servers and applications as part of their processing capabilities. An outbound packet on a network interface constitutes an event —a http reply is an event, a log entry written is an event, processing stage completed in Apache spark is an event, etc. Thus, any changes to CPU, memory, disk, database, application, services, infrastructure, IPMI, HT bus, etc, are events.

Riemann is an event stream processor with an asynchronous event model. It aggregates events and processes them with a stream language.

Figure 4 shows a typical deployment scenario, wherein Riemann is used to monitor a message queue cluster using multiple technology components.

OpenNMS

While we discuss OpenNMS architecture, an interesting feature beyond monitoring is its capability for log correlation. A Java-centric infrastructure will lean towards OpenNMS rather than Riemann.

Other monitoring tools that should be mentioned are Munin, Shinken, Sensu, Consul and Icinga2. Interested readers are welcome to explore them at leisure.

Networks, going forward

Carrier network infrastructure (CNI) and telecom operations management systems are growing at a compounded annual growth of 2.6 per cent, resulting in a US$ 2.7 trillion market. The speed of smartphone and tablet adoption, coupled with cloud infrastructure and video streaming, is disrupting centralised systems, driving network traffic and changing traffic patterns.

New technologies like software defined networking (SDN), network function virtualisation (NFV), 100Gbps+, Data Centre Ethernet, Packet Optical Transport Systems (P-OTS), 5G, LTE-advanced and small cell technology have the potential to change network cost structures, and improve network flexibility and resilience.

Tools and products that provide distributed network monitoring with tomography and real-time route analytics capabilities have the potential to be game changers. END

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